

Statistics Unit 4: Simple Regression Lecture Notes

Review: Similarity between variance and covariance

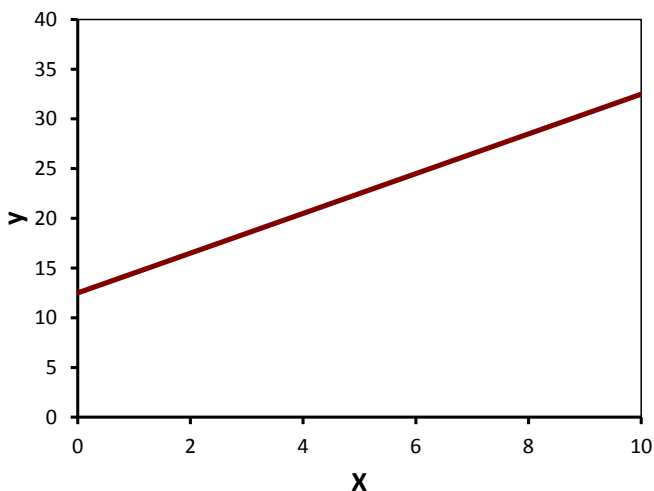
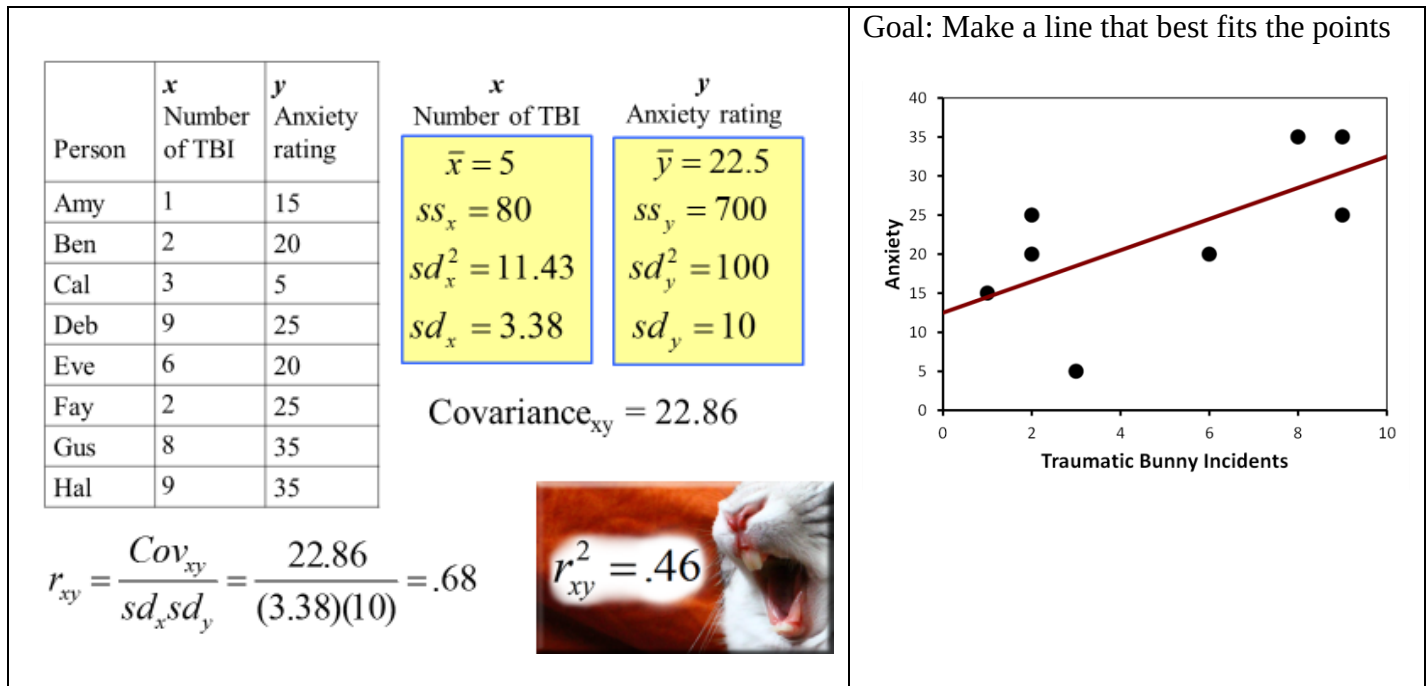
$$\text{variance} = \frac{\sum (x - \bar{x})(x - \bar{x})}{n - 1} \quad \text{covariance} = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

Correlation

$$r_{xy} = \frac{\text{Cov}_{xy}}{sd_x sd_y}$$

If both x and y are z scores (sd = 1), then. . .

Squared correlation = percent of variance explained



Math review:

$$y = 12.5 + 2x$$

For example:

If $x = 3$, then

$$y = 12.5 + (2)(3) = 18.5$$

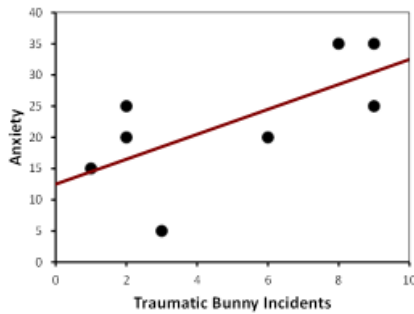
If $x = 6$, then

$$y = 12.5 + (2)(6) = 24.5$$

$$y = \text{intercept} + (\text{beta})(x)$$

$$y = b_0 + b_1x$$

The line “ $y = 12.5 + 2x$ ” best fits these points



The line provides the best prediction of y given x

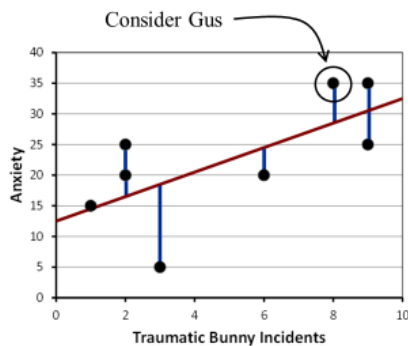
$$\hat{y} = b_0 + b_1 x \quad \text{where:} \\ b_0 = 12.5, \text{ and } b_1 = 2$$

But, the line does not perfectly predict x

The distance between y and the line is **error**

Thus, error = observed – predicted

$$= y - \hat{y}$$

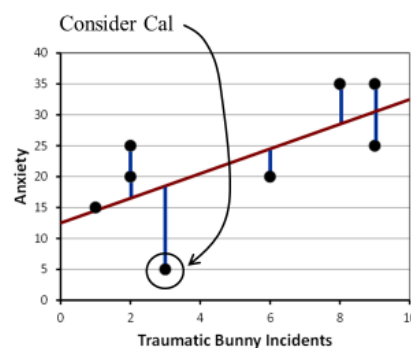


Observed
 $x = 8$
 $y = 35$

Predicted
 $\hat{y} = 28.5$

Error = 6.5

$$y = b_0 + b_1 x + e \\ 35 = 12.5 + (2)(8) + 6.5$$



Observed
 $x = 3$
 $y = 5$

Predicted
 $\hat{y} = 18.5$

Error = -13.5

$$y = b_0 + b_1 x + e \\ 5 = 12.5 + (2)(3) + (-13.5)$$

$$y = b_0 + b_1 x + e$$

b_0 = the expected value of y when $x = 0$

b_1 = the expected increase in y given a 1 unit increase in x

Estimate b_0 and b_1 such that e^2 is small as possible

Precisely, we minimize:

$$\Sigma(y - \hat{y})^2$$

Minimize the sum of squared errors

Least Squares regression

What b_1 and b_0 will minimize sum of squared error?

$$b_1 = \frac{Cov_{xy}}{sd_x^2}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

Observe the similarity between beta and correlation:

$$b_1 = \frac{Cov_{xy}}{sd_x sd_x} \quad r_{xy} = \frac{Cov_{xy}}{sd_x sd_y}$$

Person	x Number of TBI	y Anxiety rating
Amy	1	15
Ben	2	20
Cal	3	5
Deb	9	25
Eve	6	20
Fay	2	25
Gus	8	35
Hal	9	35

x Number of TBI	y Anxiety rating
$\bar{x} = 5$	$\bar{y} = 22.5$
$sd_x^2 = 11.43$	$sd_y^2 = 100$
$sd_x = 3.38$	$sd_y = 10$

$$\text{Covariance}_{xy} = 22.86$$

$$b_1 = \frac{Cov_{xy}}{sd_x^2} = \frac{22.86}{11.43} = 2$$

$$r_{xy} = .68$$

$$b_0 = \bar{y} - b_1\bar{x} = 22.5 - (2)(5) = 12.5$$

If you turn x and y into z-scores,
and the correlation between x and y is .78

What will the intercept be?

What will the slope be?

What will the covariance be?

Standardized beta: all variables turned into z-scores
(mean = 0, $sd = 1$)

$$b_1 = \frac{Cov_{xy}}{sd_x^2}$$

$$r_{xy} = \frac{Cov_{xy}}{sd_x sd_y}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

Standardized

$$= \frac{Cov_{xy}}{1} = Cov_{xy}$$

$$= \frac{Cov_{xy}}{1} = Cov_{xy}$$

$$= 0 - b_1 0 = 0$$

The full linear equations with subscripts

$$y_{is} = b_{0s} + b_{1s}x_{is} + e_{is}$$

$$b_{0sp} = \beta_{0p} + e_{0sp}$$

$$b_{1sp} = \beta_{1p} + e_{1sp}$$

Identify:

1. Equation for variance in a sample
2. Equations for sampling error variance in a population
3. Variables
4. Parameter estimates

Two key questions:

1. Descriptive

Predictive equations

$$y_{is} = \textcircled{b_{0s}} + \textcircled{b_{1s}} x_{is} + \textcircled{e_{is}}$$

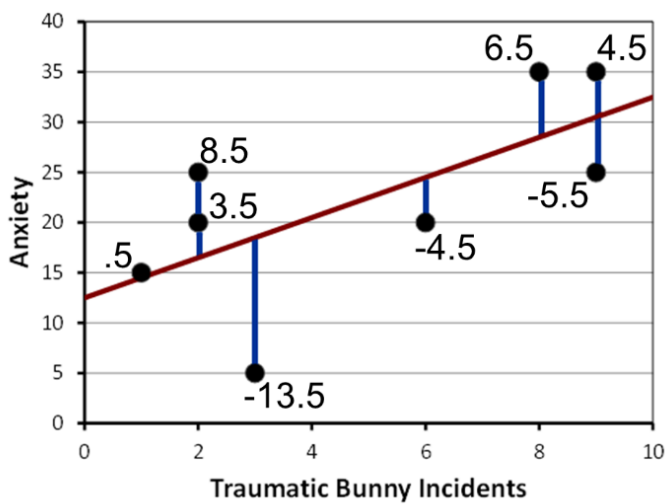
$$b_{0sp} = \beta_{0p} + e_{0sp}$$

$$b_{1sp} = \beta_{1p} + e_{1sp}$$

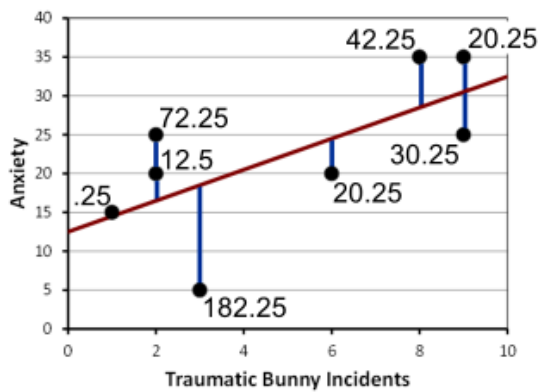
What is b_{0s} ? (not interesting)

What is b_{1s} ?

What is the average deviation of $|e_{is}|$?



Sum of errors will always = 0

Sum of squared errors

$$\begin{array}{r}
 0.25 \\
 12.25 \\
 182.25 \\
 30.25 \\
 20.25 \\
 72.25 \\
 42.25 \\
 + 20.25 \\
 \hline
 380.00
 \end{array}$$

Mean square error

$$sd_{est}^2 = \frac{\sum (y - \hat{y})^2}{n-2} = \frac{380}{8-2} = 63.33$$

Standard Error of Estimate

$$sd_{est} = \sqrt{\frac{\sum (y - \hat{y})^2}{n-2}} = \sqrt{63.33} = 7.96$$

*Standard Deviation**Standard Error of Estimate*

Total sum of squares
measures distance from
the mean

Sum of squared error
measures distance from the
regression line

$$ss = \sum (y - \bar{y})^2$$

$$ss_{est} = \sum (y - \hat{y})^2$$

Variance

Mean square error

$$sd^2 = \frac{\sum (y - \bar{y})^2}{n-1}$$

$$sd_{est}^2 = \frac{\sum (y - \hat{y})^2}{n-2}$$

Standard Deviation

Standard Error of Estimate

$$sd = \sqrt{\frac{\sum (y - \bar{y})^2}{n-1}}$$

$$sd_{est} = \sqrt{\frac{\sum (y - \hat{y})^2}{n-2}}$$

What is the relationship
between

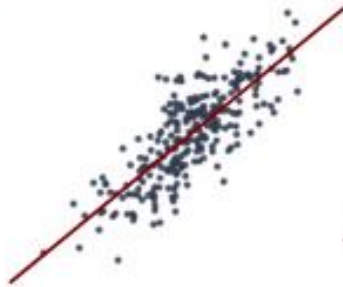
$\sum (y - \bar{y})^2$ Total sum of squares

$\sum (y - \hat{y})^2$ Sum of squared error

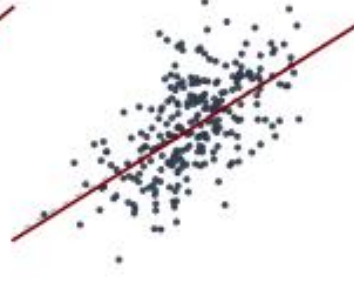
$$r = 1$$



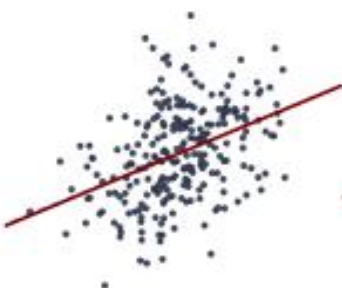
$$r = .8$$



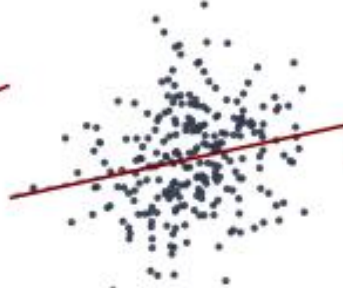
$$r = .6$$



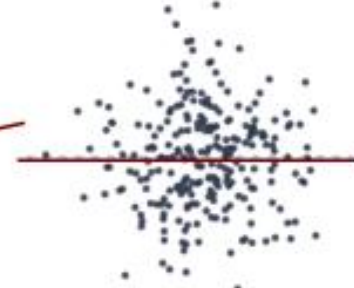
How is sum of squared error related to the slope?



$$r = .4$$



$$r = .2$$

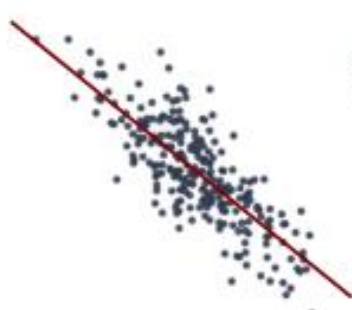


$$r = 0$$

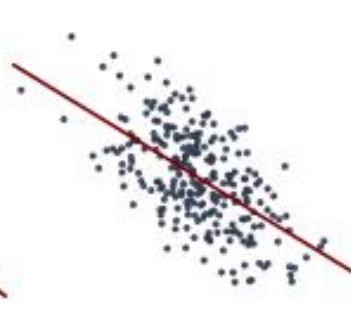
$$r = -1$$



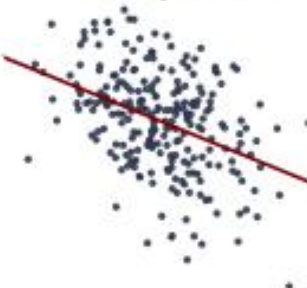
$$r = -.8$$



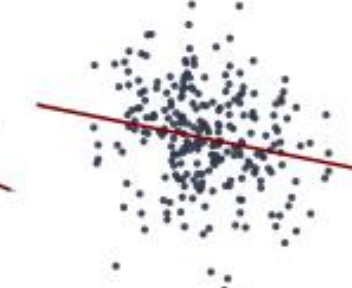
$$r = -.6$$



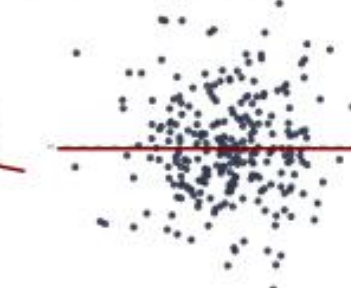
How is sum of squared error related to the slope?



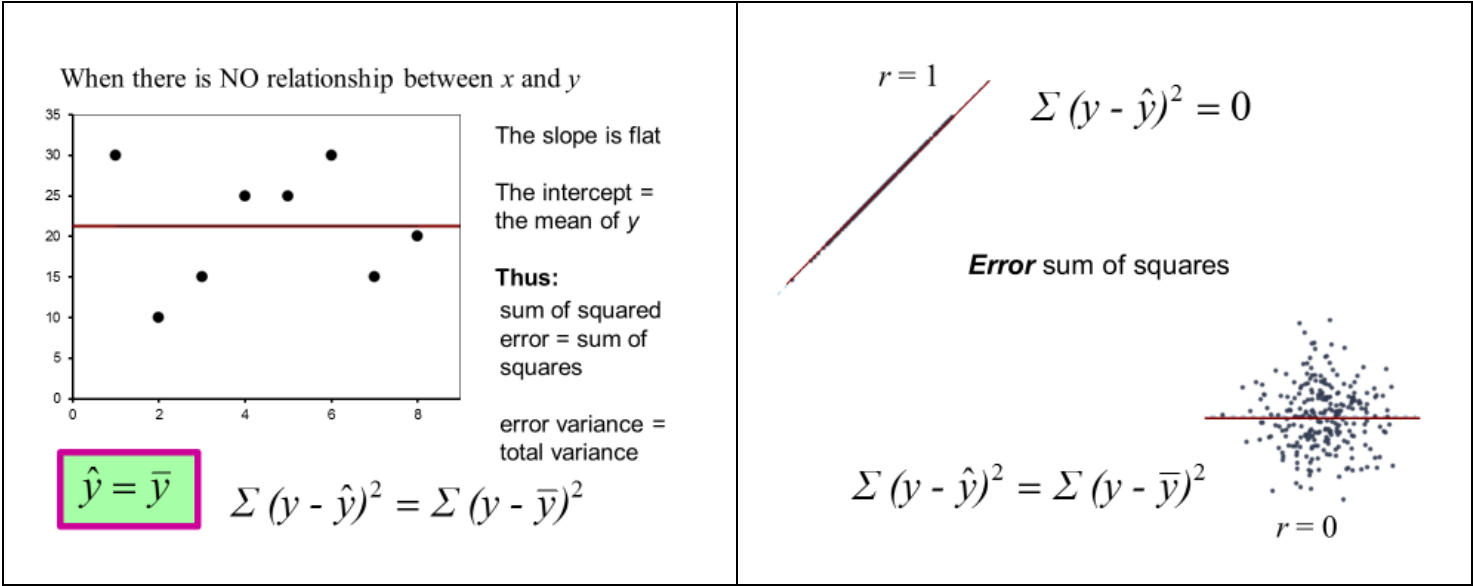
$$r = -.4$$



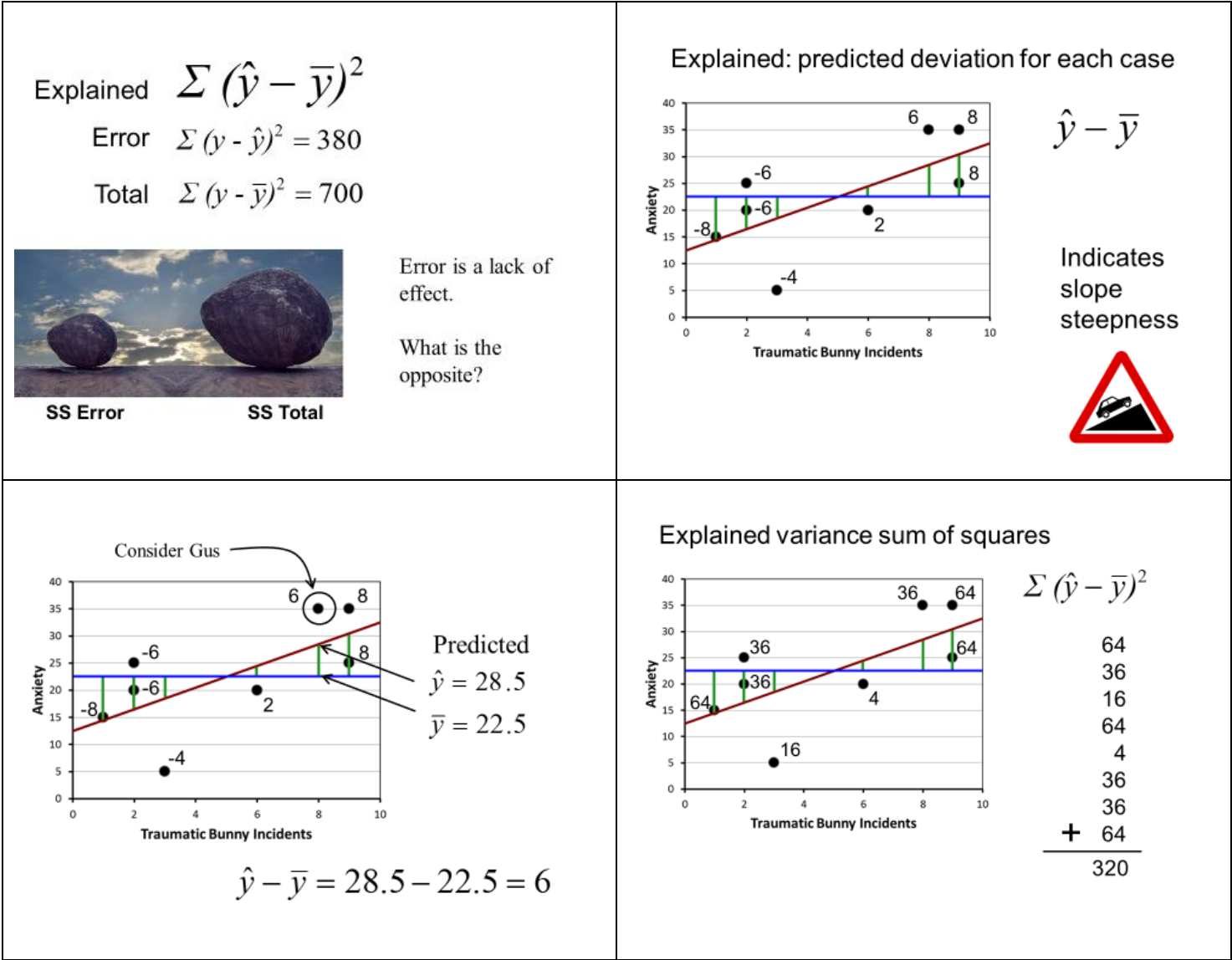
$$r = -.2$$

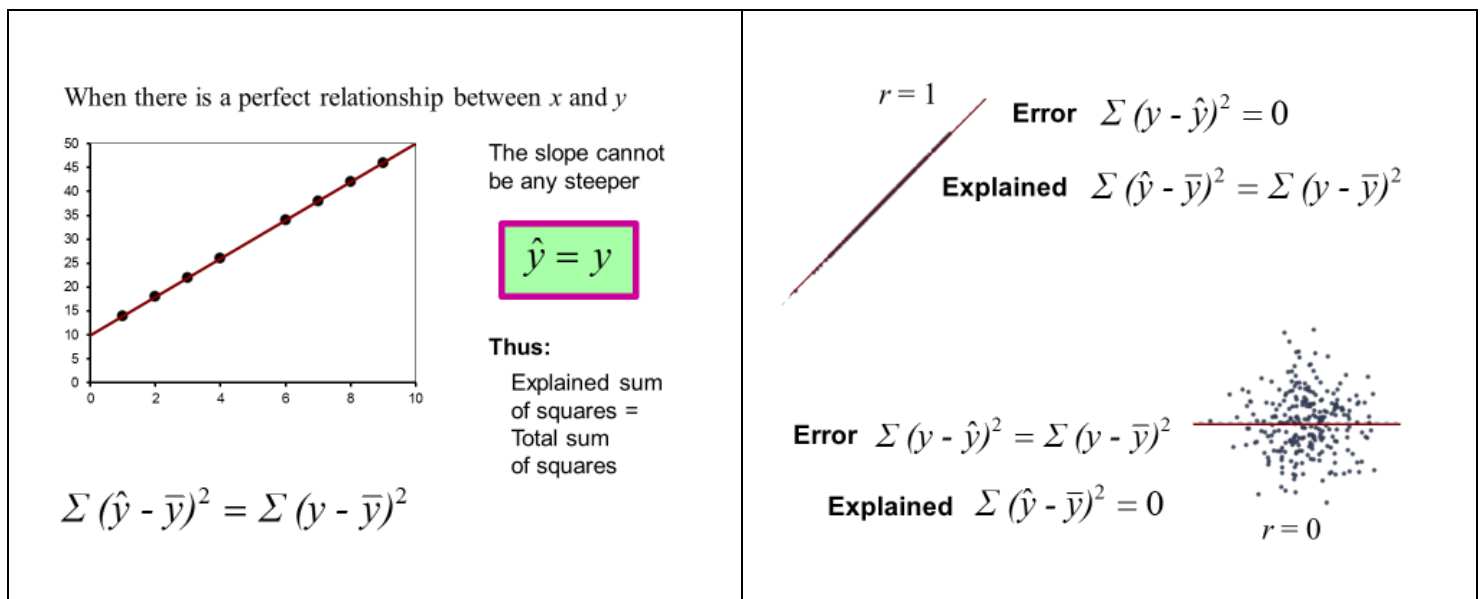
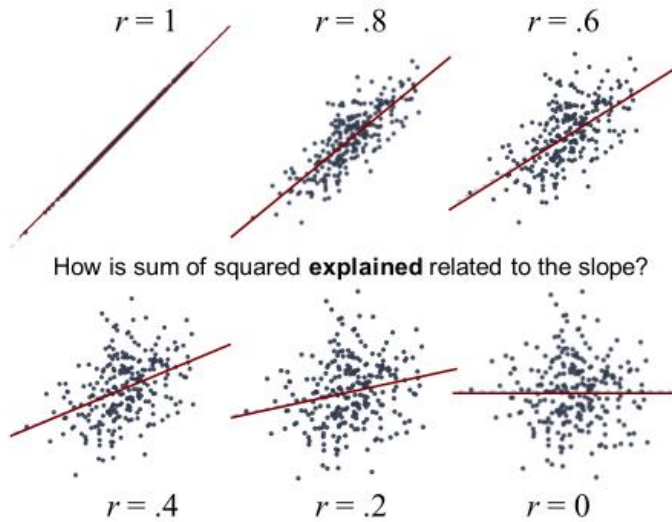


$$r = 0$$



Covariance is to variance as SS Error is to SS Total. Two different approaches, same basic idea. Sum of squared error cannot exceed total sum of squares





Sum of squares (error or explained) cannot exceed total sum of squares

Partitioning the sum of squares Explained $\sum (\hat{y} - \bar{y})^2 = 320$ Error $\sum (y - \hat{y})^2 = 380$ Total $\sum (y - \bar{y})^2 = 700$ Explained + Error = Total	Percent of variance explained $= \frac{\text{ss explained}}{\text{ss total}}$ $\frac{320}{700} = .46$ Error SS = 54% Explained SS = 46%	$r_{xy} = .68$ $r^2_{xy} = .46$
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Review:
descriptive



What is r ?

Predictive equations

$$y_{is} = b_{0s} + b_{1s} x_{is} + e_{is}$$

$$b_{0sp} = \beta_{0p} + e_{0sp}$$

$$b_{1sp} = \beta_{1p} + e_{1sp}$$

What is b_{0s} ? (not interesting)

What is b_{1s} ? (standardized $b = r$)

What is the average
deviation of $|e_{is}|$?
(standard error of estimate)



What is the percent of
variance explained?
(equal to r^2)

Two key questions:
2. Inferential

What is the average
deviation of $|e_{1sp}|$?

If H_0 is true, what is the
standardized e_{1sp} for
this study?

$$y_{is} = b_{0s} + b_{1s} x_{is} + e_{is}$$

$$b_{0sp} = \cancel{\beta_{0p}} + \cancel{e_{0sp}}$$

$$b_{1sp} = \beta_{1p} + e_{1sp}$$

Significance testing
Beta

$$sd_{beta} = \frac{sd_{est}}{sd_x \sqrt{n-1}}$$

$$t = \frac{b}{sd_{beta}}$$

Degrees of freedom = $n - 2$

Significance testing
Correlation

$$sd_r = \sqrt{\frac{1-r^2}{n-2}}$$

$$t = \frac{r}{sd_r}$$

Degrees of freedom = $n - 2$

Analysis of Variance (ANOVA)

	Sum of Squares	Degrees of Freedom	Mean Square	F
Explained (regression)	Total – Error	p	SS / df	$\frac{MS \text{ Explained}}{MS \text{ Error}}$
Error (residual)		$n - p - 1$	SS / df	
Total		$n - 1$	$p = \text{number of predictors}$	

Total sum of squares

$$ss = \sum (y - \bar{y})^2$$

Sum of squared error

$$ss_{est} = \sum (y - \hat{y})^2$$

Variance

$$sd^2 = \frac{\sum (y - \bar{y})^2}{n-1}$$

Mean square error

$$sd_{est}^2 = \frac{\sum (y - \hat{y})^2}{n-2}$$

	Sum of Squares	Degrees of Freedom	Mean Square	F
Explained (regression)	320	1	320	5.05
Error (residual)	380	6	63.33	
Total	700	7		

Predicting people's scores

$$\hat{y} = b_0 + b_1 x$$

$$b_0 = \bar{y} - b_1 \bar{x} = 12.5 \quad b_1 = \frac{Cov_{xy}}{sd_x^2} = 2$$

$$\hat{y} = 12.5 + 2x$$

Person	x Number of TBI	y Anxiety rating	$\hat{y}$
Amy	1	15	14.5
Ben	2	20	16.5
Cal	3	5	18.5
Deb	9	25	30.5
Eve	6	20	24.5
Fay	2	25	16.5
Gus	8	35	28.5
Hal	9	35	30.5

Every case has a predicted score.

$$18.5 = 12.5 + 2(3)$$

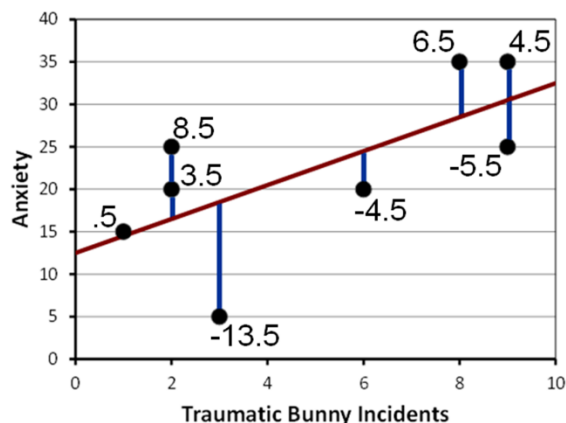
$$28.8 = 12.5 + 2(8)$$

$$\hat{y} = 12.5 + 2x$$

Every case has an error

Person	x Number of TBI	y Anxiety rating	$\hat{y}$	e
Amy	1	15	14.5	0.5
Ben	2	20	16.5	3.5
Cal	3	5	18.5	-13.5
Deb	9	25	30.5	-5.5
Eve	6	20	24.5	-4.5
Fay	2	25	16.5	8.5
Gus	8	35	28.5	6.5
Hal	9	35	30.5	4.5

$$e = y - \hat{y}$$



Extraction

How could I get a person's anxiety score with *bunny trauma* extracted out of it?

Anxiety, controlling for bunny trauma

Anxiety, holding bunny trauma constant

e = residual anxiety rating

Perpetrator and victim analogy

x variable is perpetrator, y variable is victim

x (bunny trauma) will extract all it can out of y (anxiety)

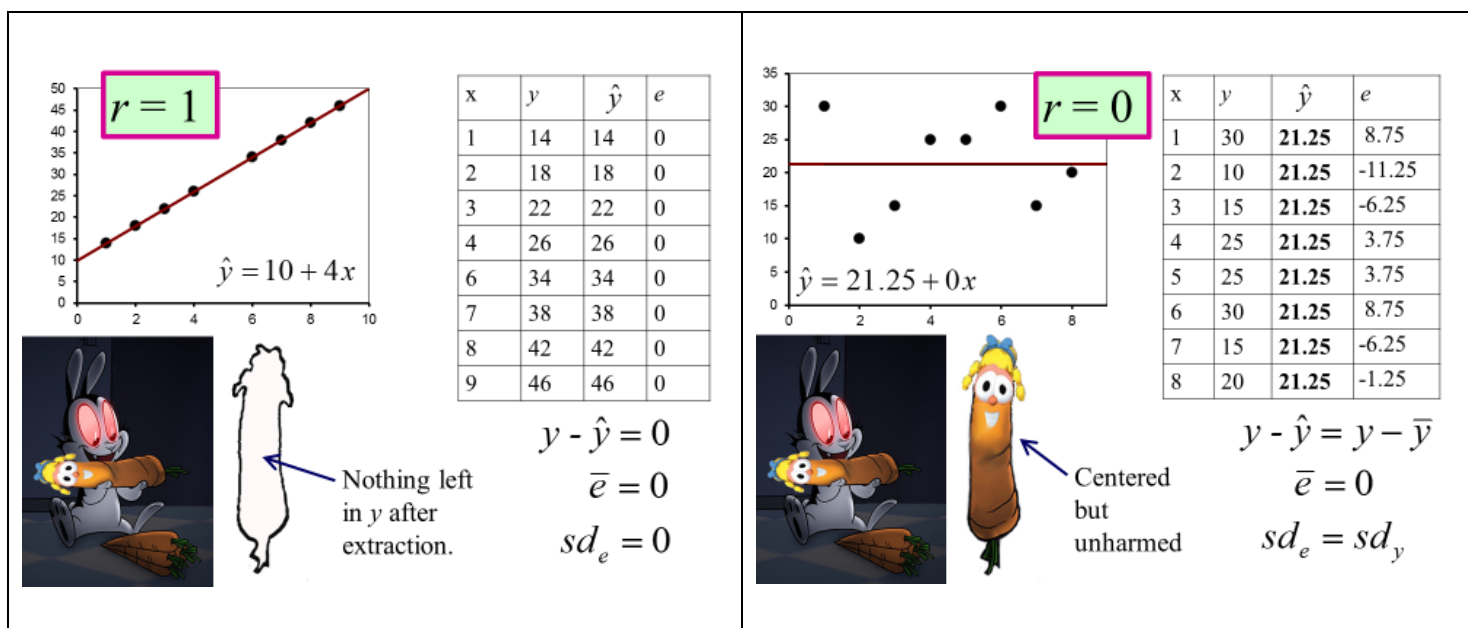
Error = Residual Score

That which is left over after x extracted all it can out of y

As the correlation increases, variance in residual scores shrinks toward zero.

The correlation between x and the residual of y will always be zero.

$$r_{x, \text{residual } y} = 0$$



Lecture Add-On Notes

Sanford, K. (2005). Attributions and anger in early marriage: Wives are event-dependent and husbands are schematic. *Journal of Family Psychology*, 19, 180-188.

77 couples, 4 conversations

Pre-conversation questionnaire

Attributions: 6 item scale, Average $\alpha = .80$

Anger: 4 Item scale, Average $\alpha = .90$

Change scores: Time 1 – Time 3

Wives

Descriptive Statistics

wife		Mean	Std. Deviation	N
1.00	ang_dif_1_3	.52	1.64	77
	att_dif_1_3	-.09	1.03	77

Inter-Item Covariance Matrix

wife		att dif 1 3	ang dif 1 3
1.00	att_dif_1_3	1.05	.92
	ang_dif_1_3	.92	2.68

Correlations

wife			att dif 1 3	ang dif 1 3
1.00	att_dif_1_3	Pearson Correlation	1	.55**
		Sig. (2-tailed)		.000
		N	77	77
	ang_dif_1_3	Pearson Correlation	.55**	1
		Sig. (2-tailed)	.000	
		N	77	77

** . Correlation is significant at the 0.01 level (2-tailed).

Model Summary

wife	Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1.00	1	.549 ^a	.302	.292	1.377

a. Predictors: (Constant), att_dif_1_3

ANOVA^b

wife	Model		Sum of Squares	df	Mean Square	F	Sig.
1.00	1	Regression	61.48	1	61.48	32.42	.000 ^a
		Residual	142.24	75	1.90		
		Total	203.72	76			

a. Predictors: (Constant), att_dif_1_3

b. Dependent Variable: ang_dif_1_3

Coefficients^a

wife	Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
			B	Std. Error	Beta		
1.00	1	(Constant)	.60	.16		3.78	.000
		att_dif_1_3	.88	.15	.55	5.69	.000

a. Dependent Variable: ang_dif_1_3

Husbands

Descriptive Statistics

husband		Mean	Std. Deviation	N
1.00	ang_dif_1_3	-.40	1.34	77
	att_dif_1_3	-.22	.97	77

Inter-Item Covariance Matrix

husband		att dif 1 3	ang dif 1 3
1.00	att_dif_1_3	.95	.25
	ang_dif_1_3	.25	1.81

Correlations

husband			att dif 1 3	ang dif 1 3
1.00	att_dif_1_3	Pearson Correlation	1	.19
		Sig. (2-tailed)		.10
		N	77	77
	ang_dif_1_3	Pearson Correlation	.19	1
		Sig. (2-tailed)	.10	
		N	77	77

Model Summary

husband	Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1.00	1	.191 ^a	.037	.024	1.328

a. Predictors: (Constant), att_dif_1_3

ANOVA^b

husband	Model		Sum of Squares	df	Mean Square	F	Sig.
1.00	1	Regression	5.01	1	5.01	2.84	.096 ^a
		Residual	132.28	75	1.76		
		Total	137.29	76			

a. Predictors: (Constant), att_dif_1_3

b. Dependent Variable: ang_dif_1_3

Coefficients^a

husband	Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
			B	Std. Error	Beta		
1.00	1	(Constant)	-.34	.16		-2.19	.03
		att_dif_1_3	.26	.16	.19	1.69	.10

a. Dependent Variable: ang_dif_1_3

Questions

1. In solving a regression equation, the computer attempts to make a certain value as small as possible. In the printed results, what number is this minimized value?
2. What is the typical distance that people's scores on the outcome variable deviated from the mean?
3. What is the typical distance that people's scores on the outcome variable deviated from the regression slope?
4. If this study were repeated over and over, how much will the estimated (unstandardized) beta typically deviate from the true population parameter?
5. If this study were repeated over and over, how much will the estimated correlation typically deviate from the true population parameter? (note: this requires a calculation)
6. What is the percent of variance that has been explained in this equation?
7. What is the expected increase in y if there is a one unit increase in x?
8. What is the expected value for y when x equals zero?
9. What is $\Sigma(y - \bar{y})^2$?
10. What is $\frac{\Sigma(y - \bar{y})^2}{n - 1}$?
11. What is $\Sigma(x - \bar{x})(y - \bar{y})$?
12. What is $\frac{\Sigma(x - \bar{x})(y - \bar{y})}{n - 1}$?
13. If Keith received a score of 0.8 on attribution change, what is What is Keith's predicted score for anger change?
14. If Keith actually received a score of -0.5 on anger change, what is the squared error for Keith's score?
15. According to the results printed above, what is the typical squared error for participants in this study?
16. Was the squared error for Keith higher or lower than the typical squared error for participants in this study?
17. According to the results printed above, what is the sum of squared errors for all participants in this study?

Is there an association between attribution change and anger change? To answer this question, provide the 5 pieces of information. When asked for the name of a statistic, type an abbreviation from the Acceptable Abbreviations Page.

18. the name of the descriptive statistic (use the Acceptable Abbreviations Page)
19. the value of the descriptive statistic
20. the name of the inferential statistic (use the Acceptable Abbreviations Page)
21. the value of the inferential statistic
22. the p-value (**report exactly as printed with NO rounding**)
26. The association is best described as:
 (A) not significant (B) significant but trivial (C) small (D) medium (E) large

Compute the following:

27. variance for the dependent variable
28. standard deviation for the dependent variable
29. standard error of estimate
30. regression sum of squares
31. total degrees of freedom
32. degrees of freedom for "regression"
33. degrees of freedom for the residual
34. mean square regression
35. mean square residual (error)

36. F value
37. Significance of F
38. R squared
39. beta weight
40. intercept
41. standardized beta
42. t value for the beta weight
43. significance of t for the beta weight
44. correlation
45. t value for the correlation
46. significance of t for the correlation

$$b_0 = 5.2 \quad \text{Sum of squares total}$$

$$b_1 = .4 \quad sd^2(df) = 2.3022^2(4) = 21.20$$

$$\text{mean of } y = 8.4$$

$$sd \text{ of } y = 2.3022$$

x Likes Peas	intercept	slope	$\hat{y}$	$\hat{y} - \bar{y}$	$(\hat{y} - \bar{y})^2$
3	5.2	0.4	6.4	-2	4.00
7	5.2	0.4	8	-0.4	0.16
9	5.2	0.4	8.8	0.4	0.16
10	5.2	0.4	9.2	0.8	0.64
11	5.2	0.4	9.6	1.2	1.44

Sum of squares explained

Percent explained

$$\sum(\hat{y} - \bar{y})^2 = 6.4 \quad \frac{6.4}{21.20} = .302$$

Sum of squares Explained

Sum of squares total

$$21.20 - 14.8 = 6.4 \quad sd^2(df) = 2.3022^2(4) = 21.20$$

	x Likes peas	y Likes pigs	Predicted y	e (residual)	e^2
Cal	3	6	6.4	-0.4	0.16
Deb	7	9	8	1	1
Eve	9	7	8.8	-1.8	3.24
Fay	10	12	9.2	2.8	7.84
Gus	11	8	9.6	-1.6	2.56

Sum of squares error

Percent explained

$$\sum e^2 = 14.8 \quad \frac{6.4}{21.20} = .302$$